QE Projects Summary

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Outline

1. Identification and Optimal Instruments in Production Function Estimation
2. Improving the Measurement of Productivity Dispersion and Misallocation in Developing Countries
3. Can Bureaucrats Solve NP-Complete Problems (and Do They Even Want To)

Identification and Optimal Instruments in Production Function Estimation

Main Project

Outline

- Motivation: Why Estimate the Production Function?
- Literature: Two Footnotes in ACF (2015)
- Model: ACF (2015)
- Theoretical Results: Polynomial Root Solver, Optimal Instruments
- Empirical Results: Point Identification in ACF Monte Carlos and Compustat
- Next Steps: Please critique these!

Motivation I: Why estimate production functions?

Object of Interest	Examples
Productivity	De Loecker (2013) Collard-Wexler & De Loecker (2015)
Marginal costs and pass-through	De Loecker et al. (2016) Burstein, Carvalho & Grassi (2025)
Markups and markdowns	De Loecker, Eeckhout & Unger (2020) Yeh, Macaluso & Hershbein (2022)

Literature I: Two Footnotes to ACF (2015)

- Framework: Ackerberg, Caves, and Frazer (2015) provided a framework for identification of production coefficients from value added production functions with invertible intermediate input demand functions.
- Footnote 16: *...there is a "global" identification issue in that the moments have expectation zero not only the true parameters, but also at one other point on the boundary of the parameter space where* $\beta_k = 0$*...*
- Footnote 11: *...Note that there are additional conditional moment conditions implied by the model related to inputs further in the past. Whether these are used as additional moments is typically a matter of preference.*

Literature II: Identification

- Point Identification: De Loecker and Syverson (2021) repeat Footnote 16's concern of spurious global minimum.
- Kim, Luo, and Su (2019) provide a practical heuristic to avoid a spurious corner solution during search: add more lagged instruments, *à la* Footnote 11.
- Weak Identification: De la Cal Medina (2026) shows ACF can be weakly identified even at interior solutions, so that standard inference breaks down. He advances inference under weak identification.

Literature III: Instrument Selection

- Kim, Luo, and Su (2019) and De la Cal Medina (2026) are going to work with moments based on "standard" instruments, i.e., lagged values of the inputs.
- Chamberlain (1987) derived optimal instruments for conditional moment models according to asymptotic-variance-minimizing criterion. In theory, these could limit researcher discretion over instrument selection.
- Valmari (2023) is, to my knowledge, the only paper to ever derive optimal instruments in the PFE setting as part of her estimation approach.

My Contribution (So Far)

1. I reframe a canonical GMM specification as a problem of finding the **roots of a quartic polynomial,** allowing me to recover all roots.
2. I provide a **definitive test for the failure of point identification**, appropriate to each application's estimation specification and dataset.
3. I demonstrate in the ACF Monte Carlo DGPs and in Compustat's manufacturing sector that the **multiple root problem includes interior roots,** suggesting non-identification cannot be solved by *a priori* parameter domain restrictions.
4. I illustrate in these environments that **point identification can depend on instrument selection decisions**.
5. To circumvent this arbitrariness, I **derive optimal instruments,** and I show that these differ materially from naive instruments. The instruments specified in the ACF appendix come closest to the optimal instruments.
6. However, I also show that the **interior root problem can be made worse by these optimal instruments.**

Model: Generic Case

Let i index the firm and t the time in a firm panel. Much literature can be understood as working within, making adjustments to, this model:

$$
\begin{aligned}
y_{it} &= f(k_{it}, l_{it}; \beta) + \omega_{it} + \epsilon_{it} \\
y_{it} &= h(k_{it}, l_{it}, m_{it}; \gamma) + \epsilon_{it} \\
\omega_{it} &= g(\omega_{i,t-1}, z_{it}; \rho) + \xi_{it} \\
0 &= E[\epsilon_{it} \mid \{k_{i\tau}, l_{i\tau}, m_{i\tau}, \omega_{i\tau}, \xi_{i\tau}, \epsilon_{i\tau-1} \text{ s.t. } \tau \leq t\}] \\
0 &= E[\xi_{it} \mid \{k_{i\tau}, l_{i\tau-1}, m_{i\tau-1}, \omega_{i\tau-1}, \xi_{i\tau-1}, \epsilon_{i\tau-1} \text{ s.t. } \tau \leq t\}]
\end{aligned}
$$

plus *a priori* parameter domain restrictions e.g. $0 < \rho < 1$.

Model: Which Specification To Study?

Paper	$F(\cdot)$	$g(\cdot)$	$h(\cdot)$	Data
Raval (2023)	CD + TL	AR(1)	$\mathrm{TP3}(k, l, m)$ + year FE	Compustat, plus six others across globe
De Ridder et al. (2026)	TL	AR(1)	$\mathrm{TP3}(k, l, m, o)$ + p, s, year FE	French FARE + EAP, 2009–2019
De Loecker et al. (2020)	CD	Nonparametric $g(\omega_{t-1})$	Nonparametric $\phi_t(v, k, z)$	Compustat; Census checks

$\mathrm{TP3}$: complete third-order polynomial; o: services; p: price; s: market share; v: COGS bundle; z: input-demand shifters.

Model: Canonical Specification

We will primarily consider the Cobb-Douglas, *m*-Leontief production case with AR(1) productivity. Instrument selection TBD, but common practice is to be just-identified:

$$
\begin{aligned}
y_{it} &= b_0 + \beta_k k_{it} + \beta_l l_{it} + \omega_{it} + \epsilon_{it} && (1)\\
y_{it} &= \gamma_0 + \gamma_l l_{it} + \gamma_k k_{it} + \gamma_m m_{it} + \epsilon_{it} && (2)\\
\omega_{it} &= \rho\omega_{i,t-1} + \xi_{it} && (3)\\
0 &= E[\epsilon_{it} \mid \{k_{i\tau}, l_{i\tau}, m_{i\tau}, \omega_{i\tau}, \xi_{i\tau}, \epsilon_{i\tau-1} \text{ s.t. } \tau \leq t\}] && (4)\\
0 &= E[\xi_{it} \mid \{k_{i\tau}, l_{i\tau-1}, m_{i\tau-1}, \omega_{i\tau-1}, \xi_{i\tau-1}, \epsilon_{i\tau-1} \text{ s.t. } \tau \leq t\}] && (5)
\end{aligned}
$$

plus *a priori* constraints that $0 < \beta_l, \beta_k, \rho < 1$.

Model: Canonical Residual Functions

Solving for residual functions

$$\epsilon_{it}(\theta) = y_{it} - \gamma_0 - \gamma_l l_{it} - \gamma_k k_{it} - \gamma_m m_{it} \tag{6}$$

$$\begin{aligned}\xi_{it}(\theta) := {} & \gamma_0 + \gamma_l l_{it} + \gamma_k k_{it} + \gamma_m m_{it} - \beta_k k_{it} - \beta_l l_{it} - (1-\rho) b_0 \\ & - \rho\Big(\gamma_0 + \gamma_l l_{i,t-1} + \gamma_k k_{i,t-1} + \gamma_m m_{i,t-1} - \beta_k k_{i,t-1} - \beta_l l_{i,t-1}\Big)\end{aligned} \tag{7}$$

Estimation: Method of Moments

Given panel data X_{IT}, let $N := I(T-1)$. Specify intuitive instrument sets.

Then $\widehat{\theta} := (\widehat{\gamma_0}, \widehat{\gamma_l}, \widehat{\gamma_k}, \widehat{\gamma_m}, \widehat{\beta_l}, \widehat{\beta_k}, \widehat{b_0}, \widehat{\rho})$ solves the following system of eight equations:

$$\begin{bmatrix} \overline{\epsilon(\widehat{\theta})} \\ \overline{\epsilon(\widehat{\theta})k} \\ \overline{\epsilon(\widehat{\theta})l} \\ \overline{\epsilon(\widehat{\theta})m} \end{bmatrix} = \mathbf{0}_4 \qquad \begin{bmatrix} \overline{\xi(\widehat{\theta})} \\ \overline{\xi(\widehat{\theta})k} \\ \overline{\xi(\widehat{\theta})l_{-1}} \\ \overline{\xi(\widehat{\theta})m_{-1}} \end{bmatrix} = \mathbf{0}_4.$$

Here, the bar denotes a sample average, and the subscript -1 denotes a lag.

Estimation: Solving For $\hat{\gamma}$

1. $(\widehat{\gamma_0}, \widehat{\gamma_l}, \widehat{\gamma_k}, \widehat{\gamma_m})$ can be solved in closed form by OLS, which implements the first four moments.
2. This is exactly the ACF first stage. It does not vary with second stage instrument choice.

Estimation: Expressing the Second Stage as a Quartic Polynomial

1. Reparameterize the remaining four moments

$$\begin{bmatrix} d_0 \\ d_l \\ d_k \end{bmatrix} := \begin{bmatrix} b_0 - \widehat{\gamma}_0 \\ \beta_l - \widehat{\gamma}_l \\ \beta_k - \widehat{\gamma}_k \end{bmatrix} \qquad \widetilde{\theta} := (d_0, d_l, d_k, \rho) \implies \begin{bmatrix} \overline{\xi(\widehat{\widetilde{\theta}})} \\ \overline{\xi(\widehat{\widetilde{\theta}})k} \\ \overline{\xi(\widehat{\widetilde{\theta}})l_{-1}} \\ \overline{\xi(\widehat{\widetilde{\theta}})m_{-1}} \end{bmatrix} = \mathbf{0}_4$$

2. ... arrive at a quartic in d_k alone where $r_0, \ldots, r_4$ depend only on data:

$$r_4 d_k^4 + r_3 d_k^3 + r_2 d_k^2 + r_1 d_k + r_0 = 0$$

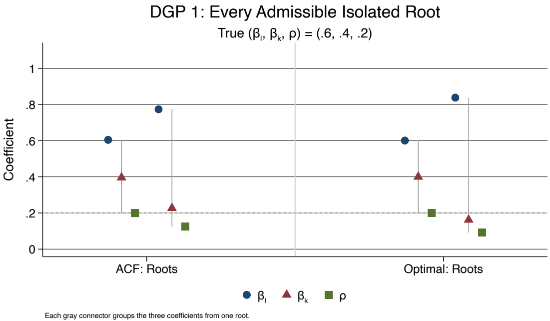
3. Each root d_k implies remaining d_l, d_0, ρ. Apply domain restrictions on $\beta' s, \rho$ to identify admissible roots.

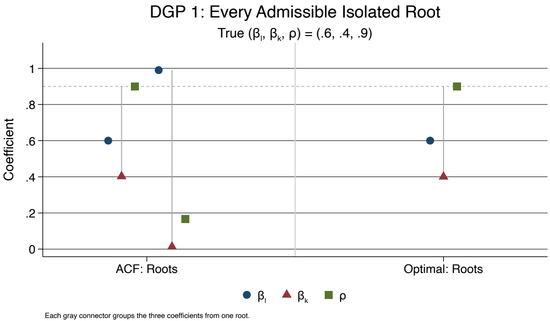
Estimation: Optimal Instruments

- Apply Chamberlain (1987) to second stage system to find optimal, specifying the set of valid instruments X_{it}.
- Each instrument choice implies its own polynomial root problem. Algebra problem is harder, not impossible, when instruments depend on parameter.

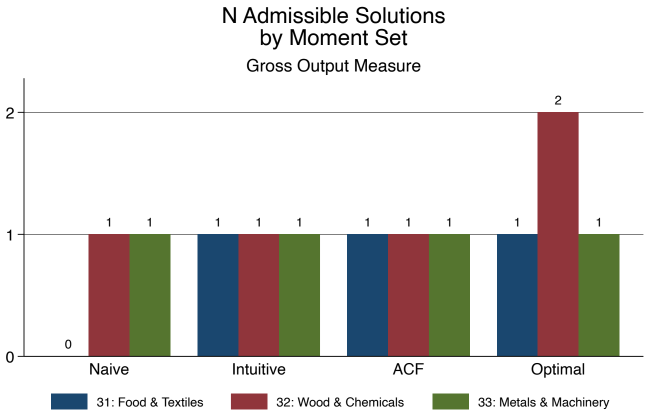
$$
\begin{array}{cccc}
\textbf{Naive} & \textbf{Intuitive} & \textbf{ACF} & \textbf{Optimal} \\
\begin{bmatrix} 1 \\ k_{it} \\ l_{it-1} \\ k_{it} l_{it-1} \end{bmatrix} &
\begin{bmatrix} 1 \\ k_{it} \\ l_{it-1} \\ m_{it-1} \end{bmatrix} &
\begin{bmatrix} 1 \\ k_{it} \\ l_{it-1} \\ \omega_{it-1}(\theta^*) \end{bmatrix} &
\begin{bmatrix} 1 \\ k_{it} - \rho^* k_{it-1} \\ \mathbb{E}[l_{it} \mid X_{it}] - \rho^* l_{it-1} \\ \omega_{it-1}(\theta^*) \end{bmatrix}
\end{array}
$$

Results: Multiple Solutions in ACF's Monte Carlo DGP



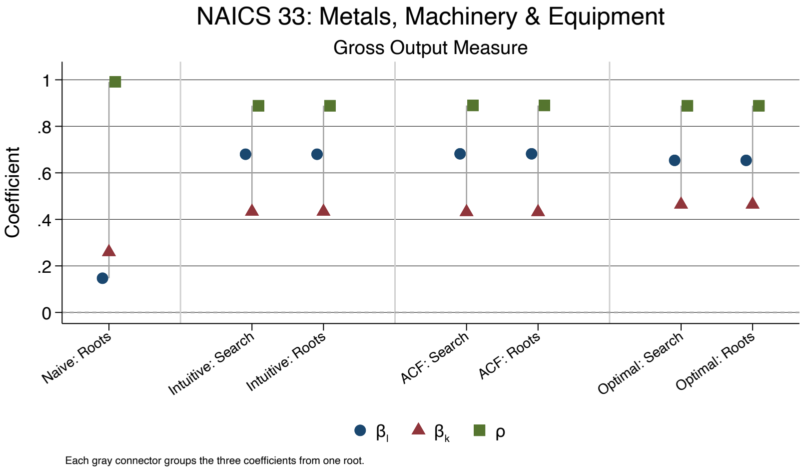


Results: Multiple Solutions in Compustat

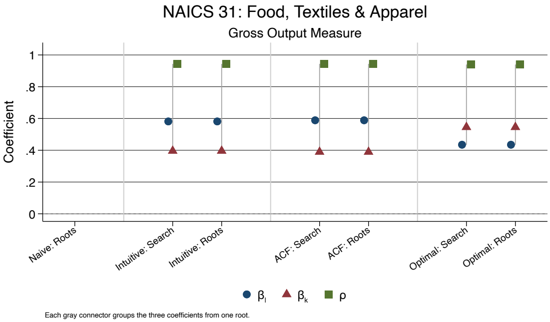


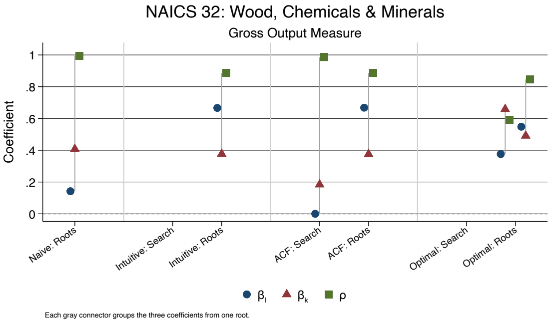
Appendix: Measured Value Added

Results: Well-Behaved Industry



Results: Problematic Industries





Possible Next Steps

From small to big:

- Demonstrate similar result in more important datasets
- Demonstrate similar result with other $f(\cdot), g(\cdot), h(\cdot)$, esp. gross value added production functions as in Gandhi, Navarro, Rivers (2020). Intentionally match more famous paper's specifications to check for spurious published results.
- Illustrate implications for downstream objects e.g. productivity, marginal cost, markup distributions.
- Resolving point identification failures is nice, but if weak identification remains, the utility of the entire approach remains limited (de la Cal Medina (2026)). Identify a minimal model that licenses residuals whose optimal instruments given both point identification and non-weak identification in canonical datasets.
- ....

Improving the Measurement of Productivity Dispersion and Misallocation in Developing Countries

Joint with Vittorio Bassi, Monica Morlacco, and Tommaso Porzio

Motivation #1: TFP Dispersion

- Fact 1: Developing countries have high variance measured TFP distributions. Measured 90-10 TFP ratios are 2:1 in the US, 3:1 in developing countries [Hsieh and Klenow (2009); De Loecker and Syverson (2021)]
- Fact 2: Interventions targeting constraints on growth for high productivity firms in developing countries often fail to produce large effects [World Bank (2019)].
- Puzzle: Why aren't high measured TFP firms benefitting from these interventions?
- Proposed Explanation: Usual TFP measurement, based on firm-level revenues and expenditures, is faulty.
- Research Question: If there is mismeasurement, what are its implications for observed TFP dispersion, characteristics of high productivity firms, and growth-constraint diagnoses?

Motivation #2: Misallocation

- Improved measurement will not eliminate measured TFP dispersion.
- Perfect competition should rule out substantial TFP dispersion, so some market failure is permitting low productivity firms to coexist alongside high productivity firms. [Mas-Colell, Whinston, and Green (1995); Hsieh and Klenow (2009)]. Recent work has documented the heightened presence of market failures in both input and output markets in developing countries [Vitali (2025); Bergquist et al (2025)]
- The result will be potentially substantial misallocation.
- Research Question: What are the sources of misallocation, and how large is it?

Proposed Contribution

Three year, three wave panel of 545 carpentry, 417 welding, 72 grain-milling firms sampled from nationally representative census in Uganda

- Firm and product-level assignment of inputs to outputs, revenue and quantity
- Direct measurement of heterogeneity (quality) of inputs and outputs
- Detailed recording of firm traits linked to TFP dispersion, market failure
    - Firm-to-firm interactions in informal clusters
    - Management and production practices
    - Access to input markets, demand

Carpentry Sample / Data

Measure	Survey wave Wave 1	Wave 2	Wave 3
Number of firms	545	518	515
Monthly total sales (UGX)	3,634,027.75	4,551,449.85	5,141,483.50
Owned capital stock, median rent-based value (UGX)	43,714.98	46,539.55	62,894.76
Monthly rented-capital expenditure (UGX)	203,764.71	188,153.47	172,388.35
Monthly labor expenditure (UGX)	479,262.88	482,494.38	573,154.37
Monthly materials expenditure (UGX)	1,495,043.09	1,838,637.07	2,328,548.35
Owns capital	0.70	0.71	0.79
Rents capital	0.83	0.85	0.81
Self-identifies as a cluster member	0.68	0.65	0.57
Number of main product lines	2.01	1.98	1.89
Has large institutional customers	0.42	0.40	0.44
Customers in the past week	1.91	1.61	1.98
Management score	10.47	10.48	10.50

Notes: Entries after the first row are unweighted means among nonmissing firm observations. Monetary variables are capped at their within-wave 95th percentiles. Main product lines are doors, beds, dining tables, and dining chairs. Means of binary variables are shares.

Method

1. Produce standard firm-level TFP estimates
2. Produce tailored, firm- and product-level TFP estimates based on production functions tailored to specific context (rent vs. own capital; incorporating input quality; permitting demand, input access, clustering to enter productivity process)
3. Compare and contrast coarse and tailored estimates: dispersion, correlates of high TFP
4. Explain firm-level TFP in terms of product-level TFPs, entries to productivity process

Can Bureaucrats Solve NP-Complete Problems (and Do They Even Want To)?

Joint with Josh Levy

Background and Motivation

- 12% of the US population receives food stamps, .4% of GDP
- Able-Bodied Adults Without Dependents (ABAWD) are subject to special work requirements, searching for work is not enough to satisfy.
- State governments may apply to waive ABAWD work requirements in areas with high unemployment rates, but they are given latitude in how they define these groups.
- States with strong technical capacity and weaker preferences for work requirements could use this latitude to exempt as many ABAWD from work requirements as possible.

Research Question(s), Literature

- Did state policymakers navigate this process “optimally" i.e. in a way that maximized the number of ABAWDs receiving waivers? [Varian (1990)]
- To the extent that policymakers didn’t optimize, is this failure better explained by a lack of technical capacity or by their underlying preferences? [Caplin and Dean (2011)]
- Which accounts of their technical constraints best explain deviations from optimal behavior? e.g. deploying sub-optimal heuristics [Caplin and Dean (2011)]
- Which accounts of their preferences best explain deviations from optimal behavior? e.g. preferences for some geographic areas or recipient types [Bjorkgren, Blumenstock, and Knight (2025)]

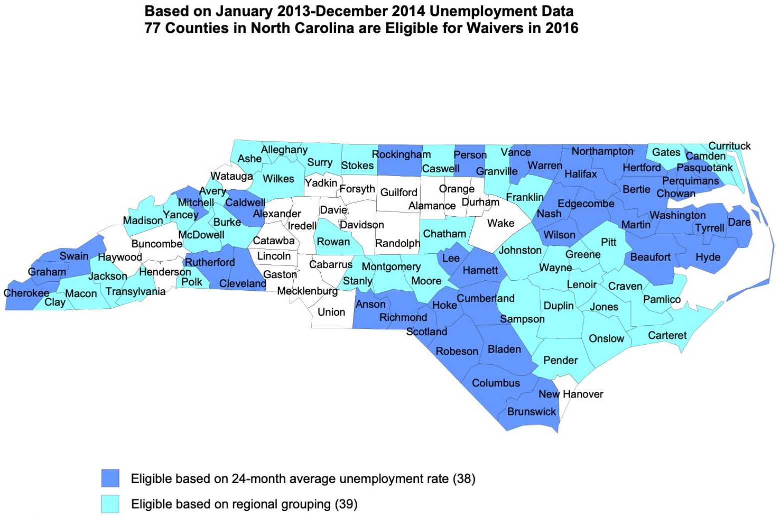
Core Policy Detail

- Any spatially contiguous set of counties qualifies for a waiver if it satisfies the following condition: there is some continuous 12-month period in the past 2 fiscal years for which the group unemployment rate was at least 20% higher than the national unemployment rate during that same 12-month period.

CALCULATING AN AVERAGE UNEMPLOYMENT RATE FOR AN AREA

1. Obtain monthly labor force data for the period (12 months, 3 months, or the historical seasonal rate)
2. Obtain monthly unemployment data for the same period
3. Total the monthly labor force data.
4. Total the monthly unemployment data.
5. Divide the unemployment total by the labor force total.
6. If the quotient has more than four decimal places, drop the fifth and all subsequent decimal places.
7. Multiply the quotient by 100. The result is the unemployment rate.

Example: North Carolina 2016



Decision Model

- Fix one 24-month window in state s
- Given county labor force and unemployment counts during this window $\{L_c, U_c\}_{c \in C_s}$ and national unemployment threshold $\alpha \in (0, 1)$
- Given the value to the decisionmaker of waiving each county $\{V_c\}_{c \in C_s}$
- Select a set of spatially disjoint, admissible county groups

$$\max_{G \in \mathcal{P}(\mathcal{P}(C_s))} \sum_{g \in G} \sum_{c \in g} V_c$$

(where $\mathcal{P}$ denotes the power set) subject to

$$g \text{ spatially contiguous } \forall g \in G \tag{8}$$

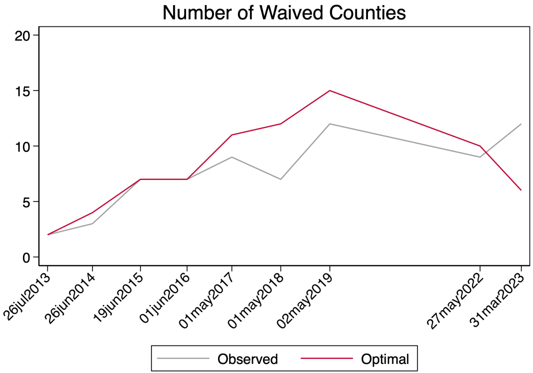
$$g, g' \in G \implies g \cap g' = \emptyset \tag{9}$$

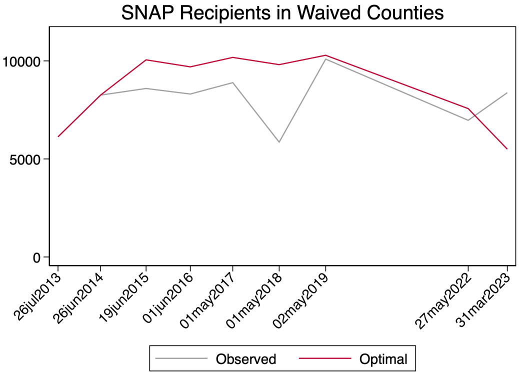
$$\frac{\sum_{c \in g} U_{ct}}{\sum_{c \in g} L_{ct}} \geq \alpha_t \; \forall g \in G \tag{10}$$

Methods

- Let V_c^{SNAP} be the number of SNAP recipients in county c, measured with Census estimates. We'll consider other valuation schemes too e.g. $V_c^N = 1 \forall c$
- Gurobi finds the solution $G^*(V)$ for each valuation scheme we propose.
- Given valuation scheme $\{V_c\}_{c \in C_s}$:
  - Value the optimal waiver allocation $G^*(V)$, giving $W_t^*(V) := \sum_{g \in G^*(V)} \sum_{c \in g} V_c$
  - Value the observed allocation from the application data G_t^{obs} giving $W_t^{obs}(V) := \sum_{g \in G_t^{obs}} \sum_{c \in g} V_c$

Preliminary Results: North Dakota





Thank you!

Appendix I: Bassi et al. (forthcoming) DGP

Bassi et al. (forthcoming) DGP

Appendix I: Model Timing I

1. The firm's productivity follows an AR(1) process from the prior period

$$Z_{it} = \mu_Z + \rho Z_{i,t-1} + \xi_{it}$$

for $\xi_{it} \stackrel{\text{iid}}{\sim} F_\xi$. The firm also draws a rental rate of capital

$$r_{it} = \mu_r \cdot \epsilon^r_{it}$$

for $\epsilon^r_{it} \stackrel{\text{iid}}{\sim} F_r$ and inherits its dynamic owned capital stock K_{it}.

2. Firm observes state (r_{it}, Z_{it}, K_{it})

3. Firm solves static optimization prior to observing ϵ_{it} but under the assumption $\epsilon_{it} \stackrel{\text{iid}}{\sim} F_\epsilon$:

$$\max_{L_{it}, C_{it} \geq 0} \mathbb{E}_{F_\epsilon}\left[e^{Z_{it}+\epsilon_{it}}(\eta K_{it}^\sigma + (1-\eta)C_{it}^\sigma)^{\frac{\beta_k}{\sigma}} L_{it}^{\beta_l} - wL_{it} - r_{it}C_{it}\right]$$

and has expected pre-investment profits $R^*(r, Z, K)$

Appendix I: Model Timing II

4. Firm chooses an investment level that solves the dynamic problem:

$$V(r,Z,K) = \max_I R^*(r,Z,K) - \mathbb{I}\{I>0\}(f_1+f_2 I^{\lambda_1}K^{\lambda_2}) - \mathbb{I}\{I<0\}(f_1+f_2 I) + \Delta \mathbb{E}_{r',Z'|Z}\left[V(r',Z',K')\right]$$

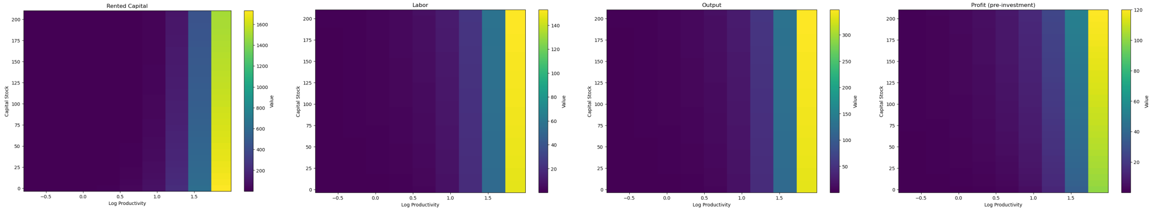
   to determine next-period capital

$$K_{i,t+1} = (1-\delta)K_{it} + I^*(r_{it}, Z_{it}, K_{it})$$

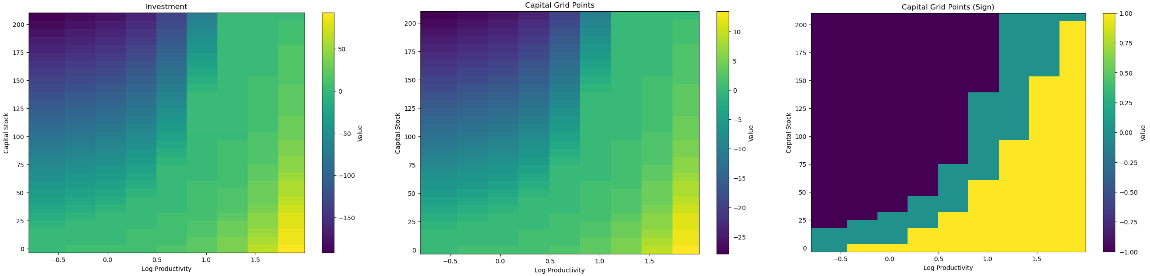
5. Firm realizes idiosyncratic productivity shock ϵ_{it} that determines final output:

$$Y_{it} = e^{Z_{it}+\epsilon_{it}}(\eta K_{it}^{\sigma} + (1-\eta)C_{it}^{*\sigma})^{\frac{\beta_k}{\sigma}} L_{it}^{*\beta_l}$$

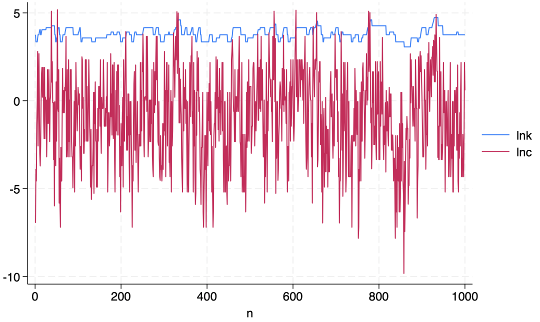
Appendix I: Simulation: Static Inputs/Output Policies

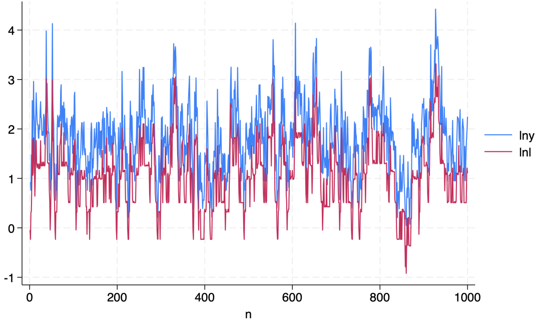


Appendix I: Simulation: Owned Capital Policies

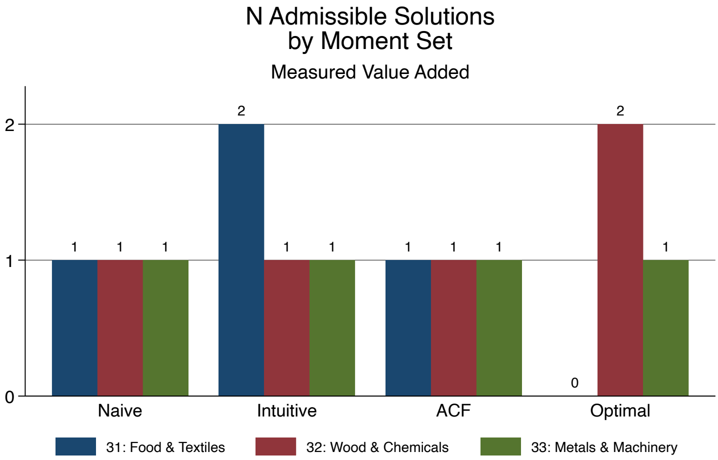


Appendix I: Simulation: Time Series

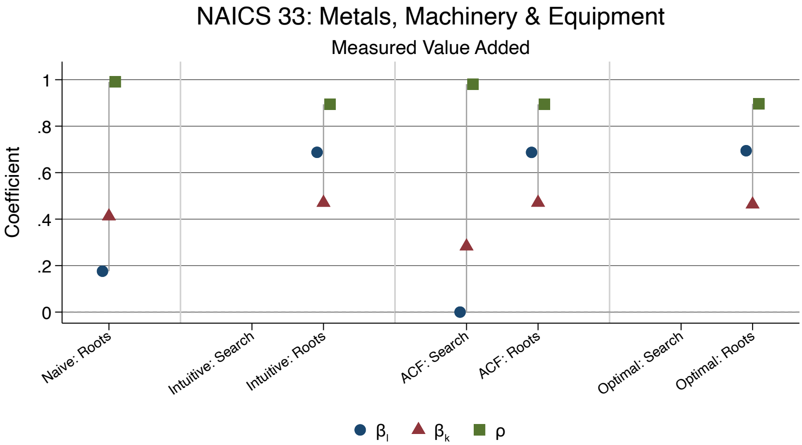




Appendix II: Measured Value Added: Multiple Solutions



Appendix II: Measured Value Added: Well-Behaved Industry



Each gray connector groups the three coefficients from one root.

Appendix II: Measured Value Added: Problematic Industries